



THARUN SAI MOKKA

Nationality: Indian **Date of birth:** 11 Dec 2001

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Home: 3-12/1, SRI SAI RAM TRACTOR MECHANICAL WORKS MAIN ROAD
KUSUMANCHI, 507159 KHAMMAM (India)

WORK EXPERIENCE

Satyam Venture Engineering Services Pvt. Ltd. – Hyderabad, India

Graduate Engineer Trainee

[3 Feb 2025 – 28 Feb 2026]

- Trained in CATIA V5 (Part Design, Surface Modelling, Assembly, Drafting) and GD&T fundamentals.
- Currently developing **automation tools for CATIA** using **Python** and **Artificial Intelligence** to enhance design efficiency and reduce repetitive tasks.
- Collaborating with cross-functional engineering teams to integrate AI-driven solutions for **parametric modeling** and **design optimization** workflows.
- Supporting continuous improvement initiatives in digital design and process automation.

Garuda 3D – Hyderabad, India

Intern

[1 Dec 2023 – 1 Jan 2024]

- Assisted in the manufacturing and assembly of 3D printers and parts, ensuring high-quality standards.
- Operated and maintained 3D printing equipment, including troubleshooting and minor repairs.
- Collaborated with the engineering team to design and prototype new components using CAD software.

DLN Engineering Systems Pvt. Ltd. – Hyderabad, India

Design Engineer (Intern)

[12 Jun 2023 – 10 Jul 2023]

- Assisted in the design and installation of HVAC systems for residential and commercial buildings.
- Participated in site inspections to assess system performance and identify areas for improvement.
- Collaborated with senior engineers to develop and revise HVAC system blueprints using AutoCAD.
- Supported the maintenance and troubleshooting of existing HVAC systems to optimize functionality and energy efficiency.

KK Heavy Machinery & Services Pvt.Ltd. – Hyderabad, India

Intern

[16 May 2022 – 15 Jun 2022]

- Studied and analyzed various arc welding techniques and their applications in manufacturing.
- Assisted in the preparation and setup of welding equipment and materials for daily operations.
- Conducted hands-on welding practices under the supervision of senior welders to develop practical skills.

PROJECTS

[18 Aug 2025]

Research Publication: Development of a 3D Printed Cervical Collar using Biocompatible and Sustainable Polylactic Acid *Published in De Gruyter Brill (2025)*

- Conducted comprehensive research on sustainable **biopolymer-based cervical collars** for medical applications.
- Designed, simulated, and 3D printed the model using **PLA**, ensuring biocompatibility and structural reliability.
- Publication focused on the integration of **ergonomic design principles, FEA validation, and additive manufacturing techniques** for healthcare innovation.
- Co-authored and published in **De Gruyter Brill Journal**, highlighting the role of eco-friendly materials in next-generation medical devices.

[3 Oct 2023 – 18 Mar 2024]

Design and Development of a 3D Printed Cervical Collar Prototype

- Designed a cervical collar prototype using Fusion 360, incorporating ergonomic and structural considerations for optimal support.
- Conducted finite element analysis (FEA) with ANSYS to ensure the prototype met strength and durability requirements.
- Utilized nTop software to optimize the lattice structure of the collar for weight reduction while maintaining rigidity.
- 3D printed the final prototype using FlashForge, ensuring high precision and quality in the manufacturing process.
- Collaborated with a multidisciplinary team to test and refine the prototype, resulting in improved comfort and functionality for end-users.

[8 Apr 2023 – 9 Apr 2024]

AI-Driven Sales Forecasting for Restaurant Item Management

- Implementation of AI to predict restaurant menu item sales, optimize inventory, reduce waste, and enhance profitability through advanced machine learning techniques.

[1 Feb 2023 – 3 Aug 2024]

Design and Precision Manufacturing of Aerospace Components

- Designing and manufacture of light weight pattern (isogrid) of aerospace components with parametric designing using CNC machines.

Designed an Industrial Conveyor Belt System for Optimal Material Handling

- Course project from **Automation and Robotics**

EDUCATION AND TRAINING

MASTER'S

DEGGENDORF INSTITUTE OF TECHNOLOGY [15 Mar 2026 – Current]

City: CHAM | Country: Germany | Level in EQF: EQF level 7

Master's in Applied Artificial Intelligence for Digital Production Management

Deggendorf Institute of Technology, Cham, Germany

Focused on the application of artificial intelligence in smart manufacturing and digital production systems, with exposure to automation, data-driven decision-making, and Industry 4.0 concepts. The program combines engineering, digital technologies, and production management to develop solutions for modern manufacturing environments.

Bachelor of Technology in Mechanical Engineering with minor in Artificial Intelligence and Machine Learning

VNR Vignan Jyothi Institute of Engineering and Technology [16 Oct 2020 – 5 Apr 2024]

City: Hyderabad | Country: India | Website: <https://vnrvjiet.ac.in/> | Field(s) of study: Mechanical Engineering | Final grade: 7.94 CGPA - Major Degree - Minor Degree (A.I & M.L)- 8.14 CGPA

Telangana State Board of Intermediate Education

Narayana Jr. College [12 Jun 2018 – 16 Apr 2020]

City: Hyderabad | Country: India | Website: <https://www.narayanagroup.com/> | Field(s) of study: Mathematics, Physics and Chemistry

Secondary School Certificate

Dr. HR Techno School [1 Jun 2013 – 5 Apr 2018]

City: Kodad | Country: India

HONOURS AND AWARDS

[18 Aug 2025]

De Gruyter Brill Journal Publication Research Paper: “Development of a 3D Printed Cervical Collar using Biocompatible and Sustainable Polylactic Acid”

[11 May 2023] IIT Kanpur

Mechatronics (NPTEL)

[15 May 2023] INVOK ENGINEERING SOLUTIONS

ANSYS

[4 Apr 2024] VNR Vignana Jyothi Institute

PROFICIENCY CERTIFICATE

• **GREEN ENERGY**

SKILLS

Adobe Photoshop / Design & Analysis: CATIA, ANSYS, Fusion 360, Autocad, nTop / Prototyping : 3D Printing / Programming : Machine Learning, Artificial Intelligence, C / Video Editing / Content Creation / Microsoft Excel / Adobe Premier Pro / Microsoft Powerpoint

LANGUAGE SKILLS

Mother tongue(s): Telugu

Other language(s):

English

LISTENING C1 READING C2 WRITING C1
SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Hindi

LISTENING B1 READING B1 WRITING B1
SPOKEN PRODUCTION B1 SPOKEN INTERACTION B1

German

LISTENING A1 READING A1 WRITING A1
SPOKEN PRODUCTION A1 SPOKEN INTERACTION A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Book Reading

Content Creation

Travelling